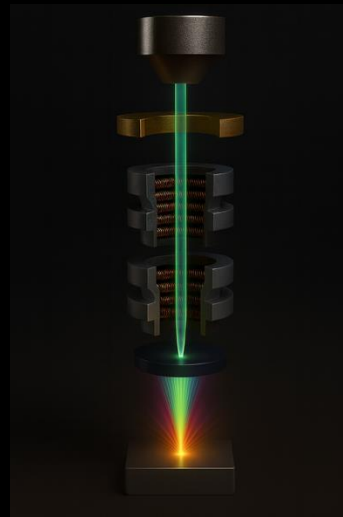
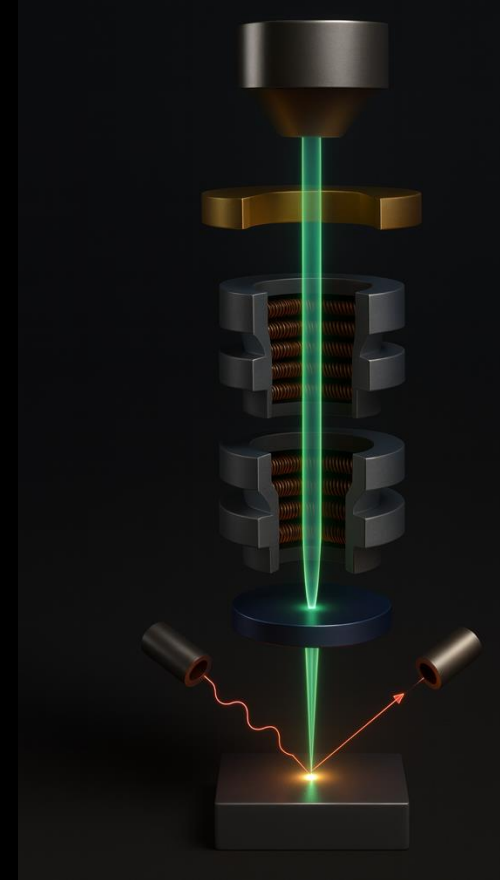
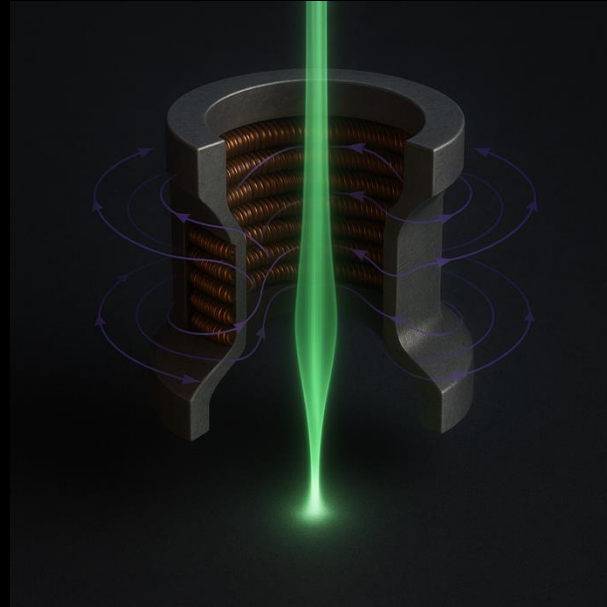
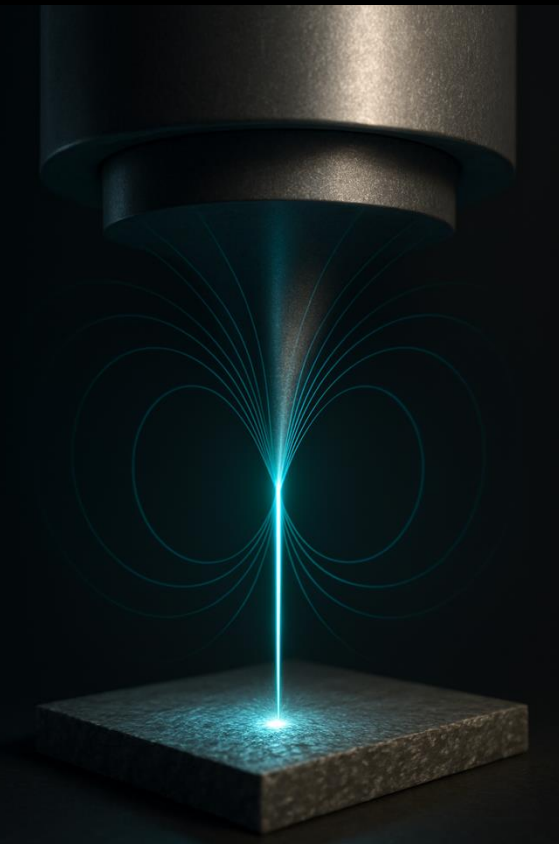
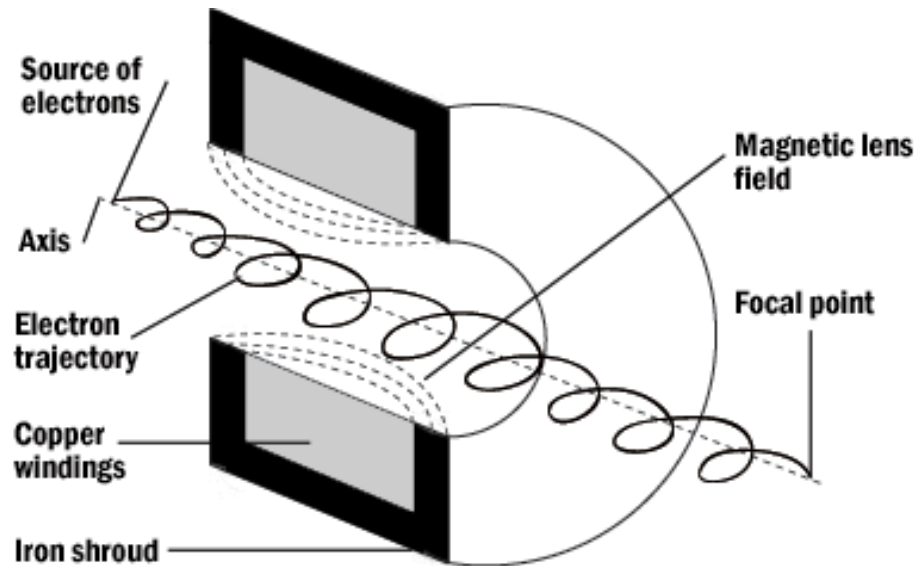


68320 Nanofabrication and Nanocharacterization Techniques

Lecture 3.5: Revision



Sample class test question



Force exerted on an electron:

$$\mathbf{F} = q \mathbf{v} \times \mathbf{B}$$

The image shows the trajectory of an electron passing through an electromagnetic lens. **Which statement is correct?**

- a) The field increases the energy of the electron.
- b) The field increases the velocity of the electron.
- c) The field increases the momentum of the electron.
- d) The field does not change the electron energy.
- e) The field decelerates the electron.

Lorentz force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$
$$= \mathbf{F}_E + \mathbf{F}_B$$

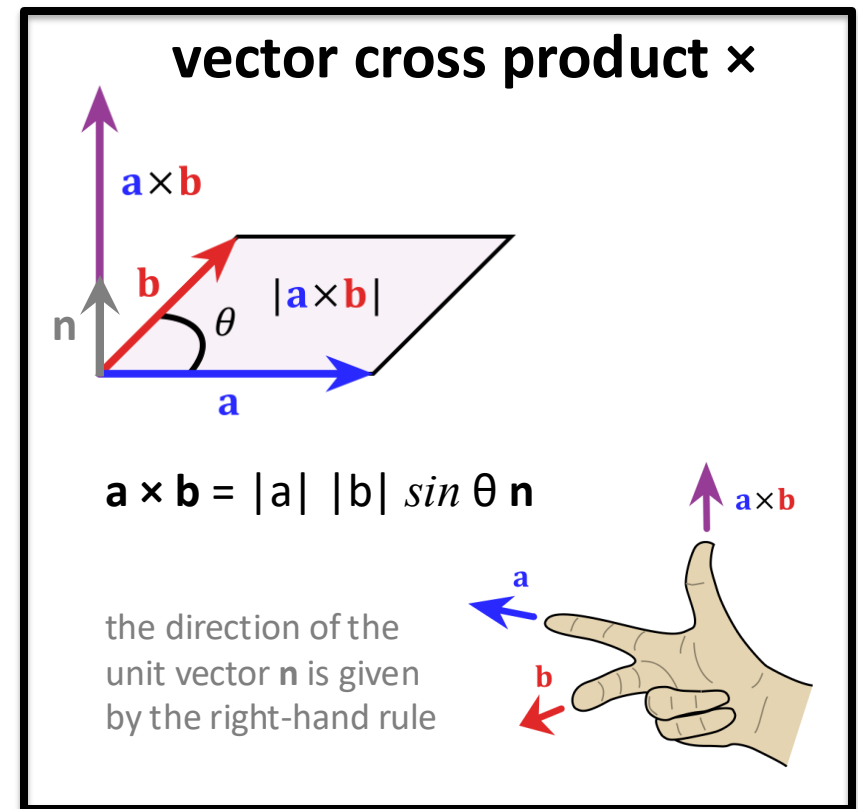
- $\mathbf{E}$ = electric field [units: $\text{NC}^{-1} = \text{Vm}^{-1}$]
- $\mathbf{B}$ = magnetic field [$\text{T} = \text{NC}^{-1}\text{m}^{-1}\text{s}$]

- $q\mathbf{E} = \mathbf{F}_E$ = electric force [N]
- $q\mathbf{v} \times \mathbf{B} = \mathbf{F}_B$ = magnetic force [N]
- $\mathbf{F}$ = “Lorentz force” [N]

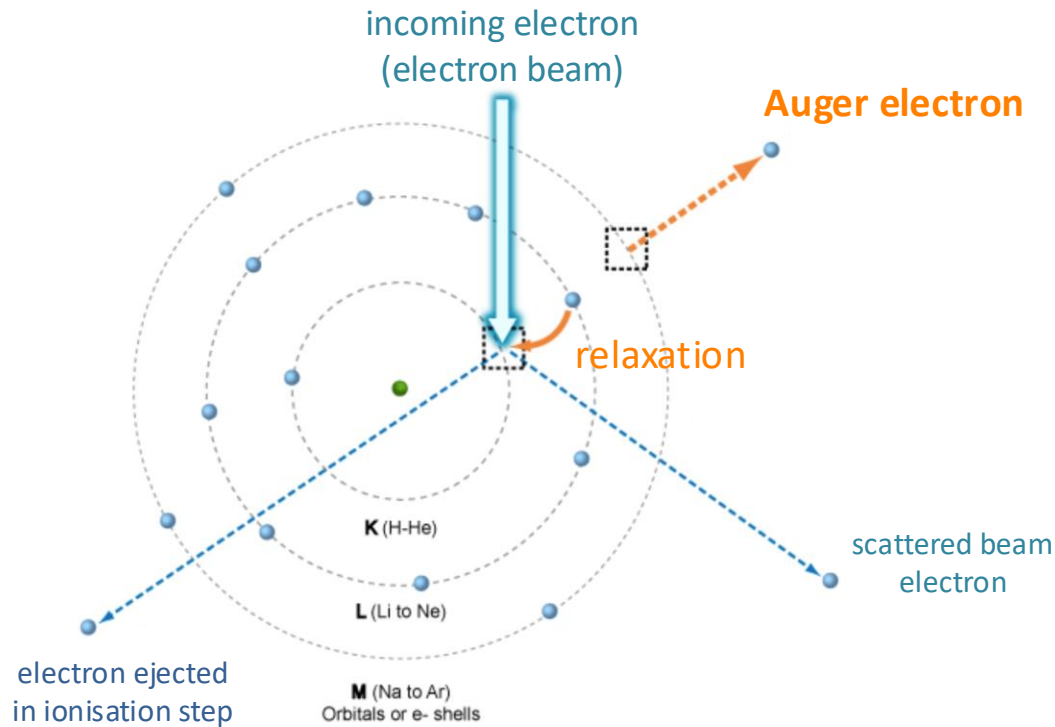
- if $q = 0$, then $\mathbf{F}_E = \mathbf{F}_B = 0$ (no force unless particle is charged)
- if $\mathbf{v} = 0$, then $\mathbf{F}_B = 0$ (no magnetic force unless particle is moving)

- Electric fields are created by charged particles & exert forces on charged particles.
- Magnetic fields are created by & exert forces on moving charged particles.
- Unlike an electric field, a constant magnetic field changes the direction of motion of a charged particle, but does not change the speed of the particle.

Typically, magnetic fields are used to focus electron beams and electric fields to focus ion beams



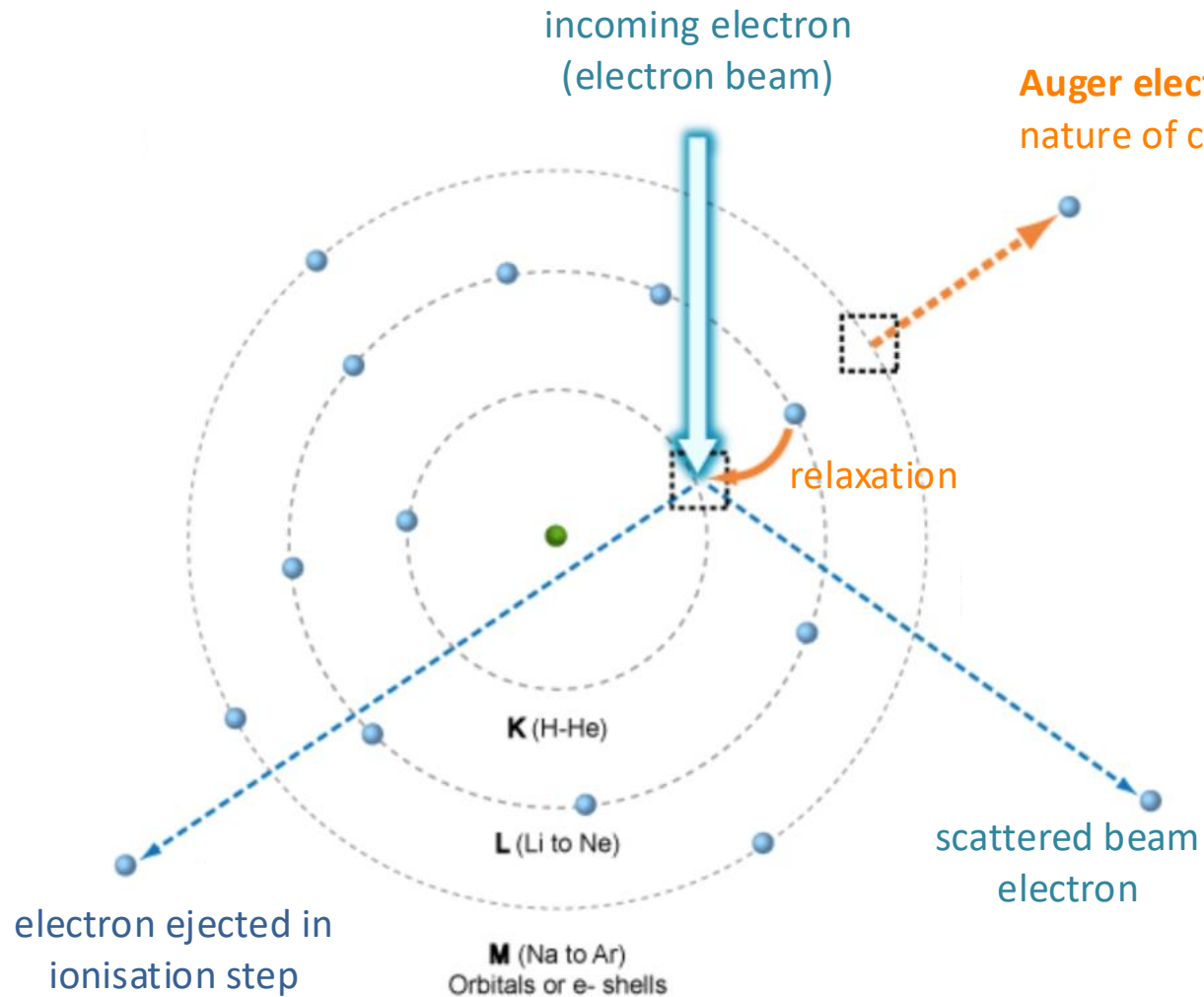
Sample class test question



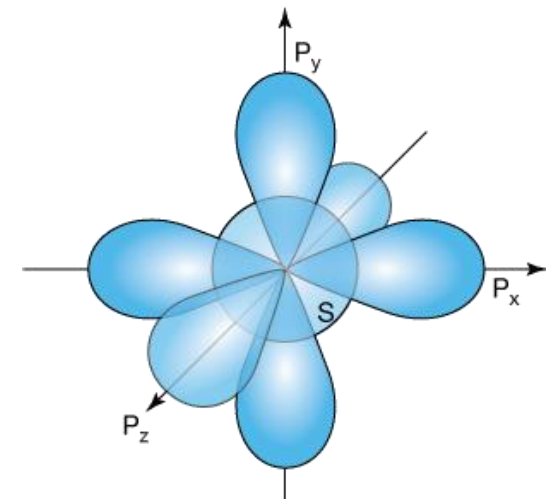
“Auger electrons provide more information on chemical bonding than x-rays emitted from a sample in an electron microscope” – Is this statement true? Why?

- a) No, because inner shell electrons participate in chemical bonds rather than outer shell electrons.
- b) No, because outer shell electrons participate in chemical bonds rather than inner shell electrons.
- c) Yes, because Auger electrons conserve energy by forming chemical bonds.
- d) Yes, because Auger electrons are entangled with nuclei in the sample.
- e) Yes, because the energy of outer shell electrons is modified by chemical bonds.
- f) Yes, because the energy of outer shell electrons is modified by neighbouring atoms.

Auger electron emission

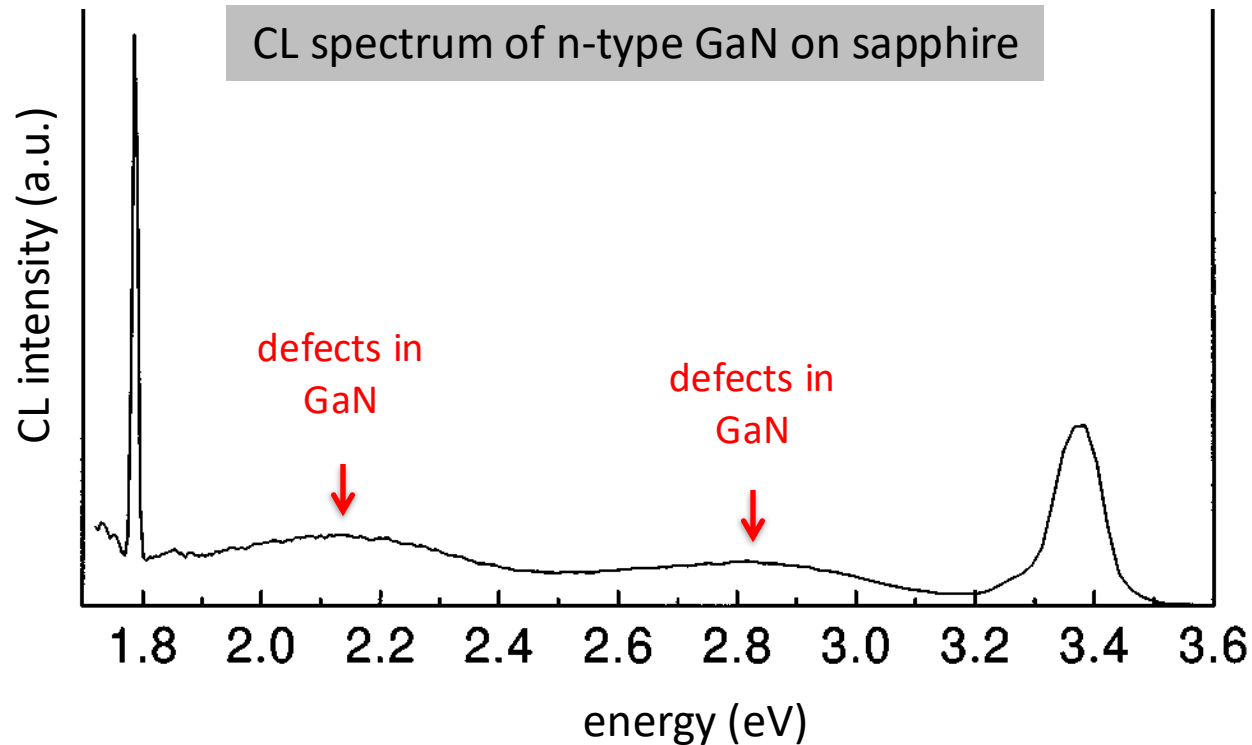


The Auger electron emission process is triggered by internal electron-electron interactions within an atom – these interactions pertain to overlaps between the electron wavefunctions within the solid



electron orbitals are representations of
the wavefunctions of electrons bound
to a nucleus

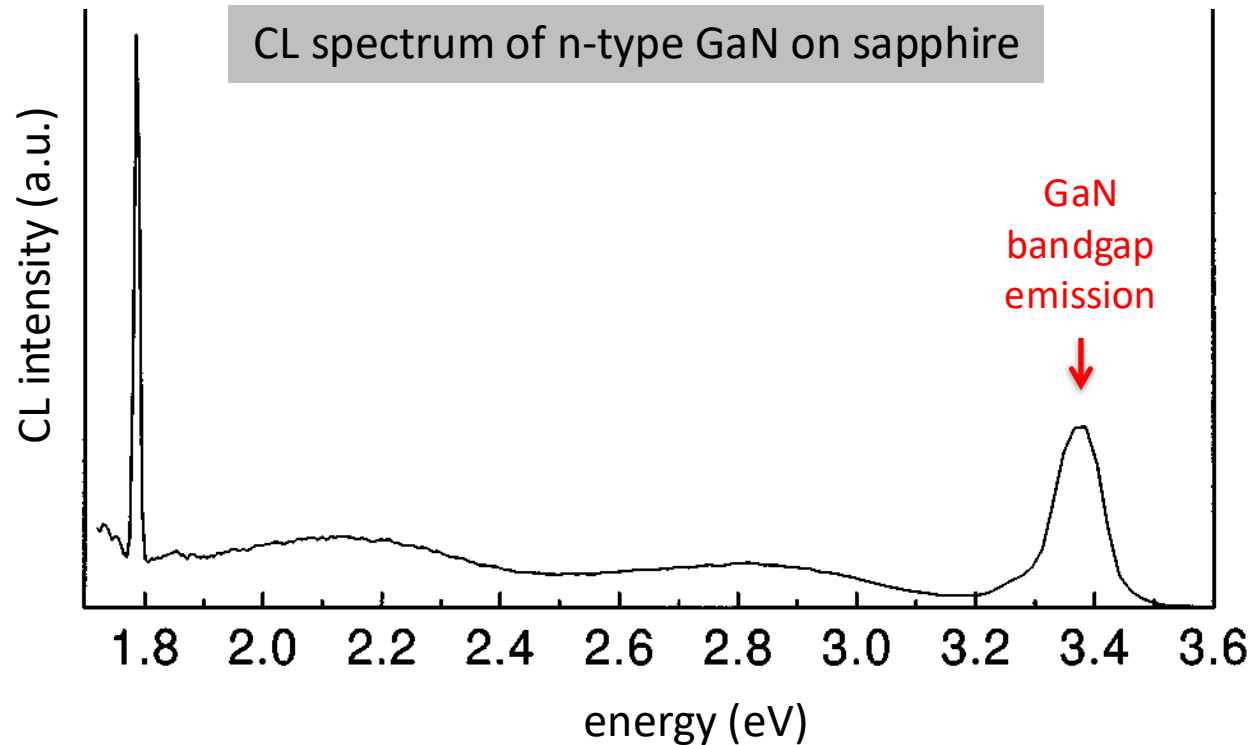
CL 1



Consider this cathodoluminescence (CL) spectrum from the semiconductor GaN. The GaN thickness is 2 microns and the electron beam energy is 30 keV. Which of the following limits the spatial resolution of CL images generated using the CL emissions indicated in red on the spectrum?

- a) Electron beam energy.
- b) Energy of the CL photons.
- c) Absorption of light by GaN.
- d) Wavelength of the CL photons.
- e) Electron-hole pair recombination rate.
- f) Atomic weight of the defects responsible for the CL emissions.
- g) Bandgap of GaN.

CL 2



Consider this cathodoluminescence (CL) spectrum from the semiconductor GaN. The GaN thickness is 2 microns and the electron beam energy is 30 keV. Which of the following limits the spatial resolution of CL images generated using the bandgap emission indicated in red on the spectrum?

- a) Electron beam energy.
- b) Energy of the CL photons.
- c) Absorption of light by GaN.
- d) Wavelength of the CL photons.
- e) Electron-hole pair recombination rate.
- f) Atomic weight of the defects responsible for the CL emissions.
- g) Bandgap of GaN.

Sample test question: Ion beam lens

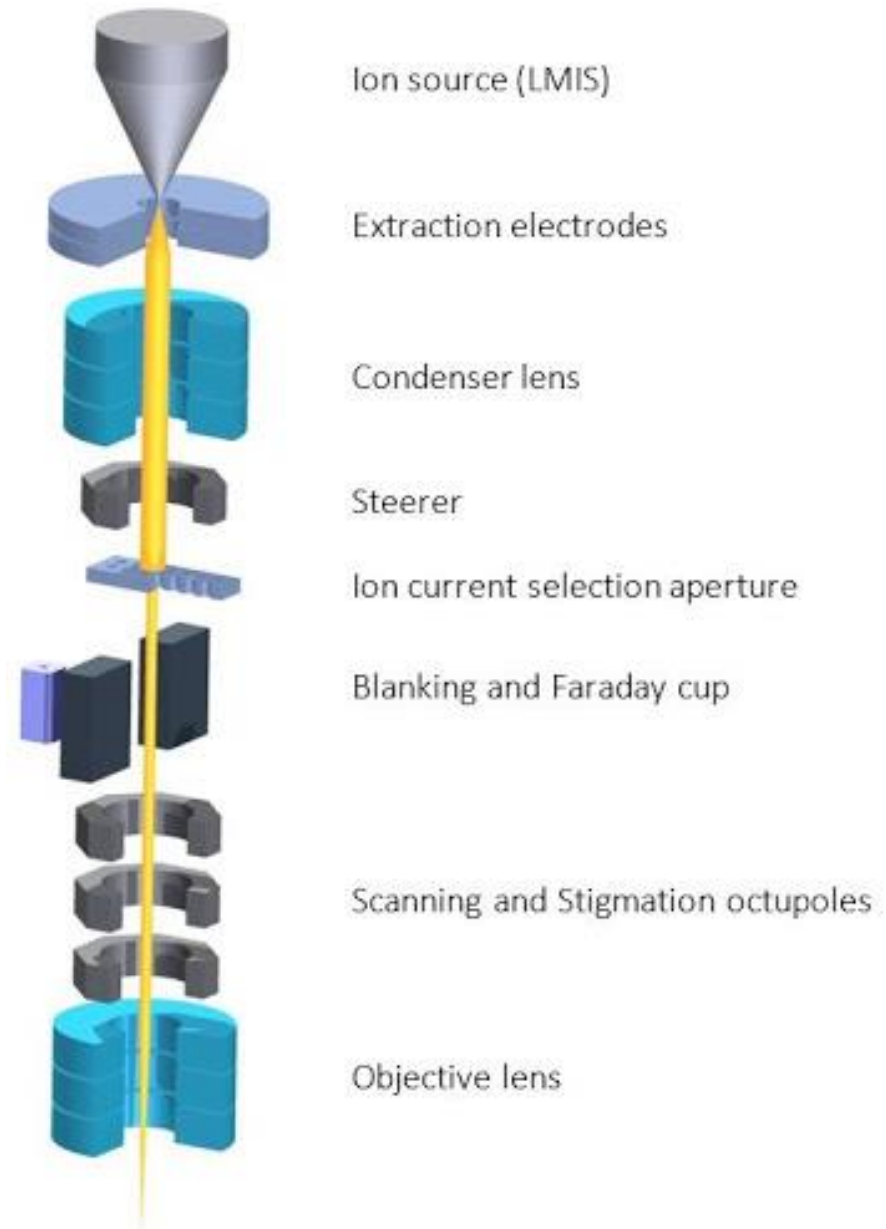
The operator of a focused ion beam (FIB) microscope increases the magnitude of the voltage applied to a 3 electrode electrostatic Einzel lens. This causes the FIB image to become blurred. Which one of these actions can be used to re-focus the image?

- a) Decrease the working distance.
- b) Increase the ion beam scan speed.
- c) Decrease the ion beam scan speed.
- d) Increase the working distance.
- e) Decrease the ion beam energy.

Sample question: Ion beam column

Consider this schematic of an ion beam column. Why does the focal length (f) of a perfect, symmetric condenser lens affect the spatial resolution of the image?

- a) Because a perfect condenser lens is designed to control the location of the focal plane below the pole piece.
- b) Because f alters the fraction of ions transmitted through the current-limiting aperture.
- c) Because a perfect condenser lens is designed to increase the energy distribution of the ions.
- d) Because a perfect condenser lens is designed to decrease the energy distribution of the ions.



Electron vs ion optics

Consider an objective lens in a scanning electron microscope, and one in a focused ion beam microscope. Which statement is true?

- a) Magnetic lenses are typically used in ion beam microscopes because the Lorentz force experienced by an ion passing through the lens is greater than that experienced by an electron passing through the lens.
- b) Electric lenses are typically used to focus electron beams because electric fields do not modify the velocity of electrons.
- c) Magnetic lenses are typically used to focus ion beams because magnetic fields accelerate ions.
- d) Magnetic lenses are typically used to focus electron beams because magnetic fields do not accelerate electrons.

Lorentz force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$
$$= \mathbf{F}_E + \mathbf{F}_B$$

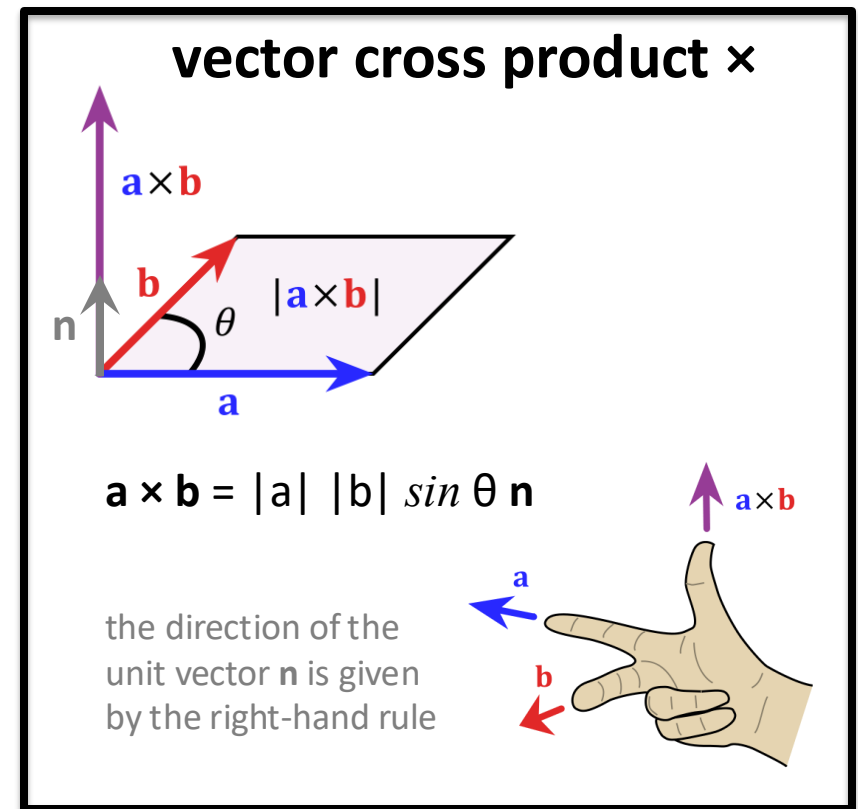
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Typically, magnetic fields are used to focus electron beams and electric fields to focus ion beams



Electron beam range (R_e) and energy (E_o)

You are using an electron microscope to perform elemental x-ray analysis of a sample that contains SiO_2 , Pt and Fe impurities. This equation describes the electron beam range R_e in the sample, as a function of the electron beam energy E_o . Which statement is true?

The diagram illustrates the factors influencing the electron beam range R_e and its physical representation. On the left, the equation for R_e is presented in a yellow box:

$$R_e(\text{nm}) = \frac{27.6 A E_o^{5/3}}{Z^{8/9} \rho}$$

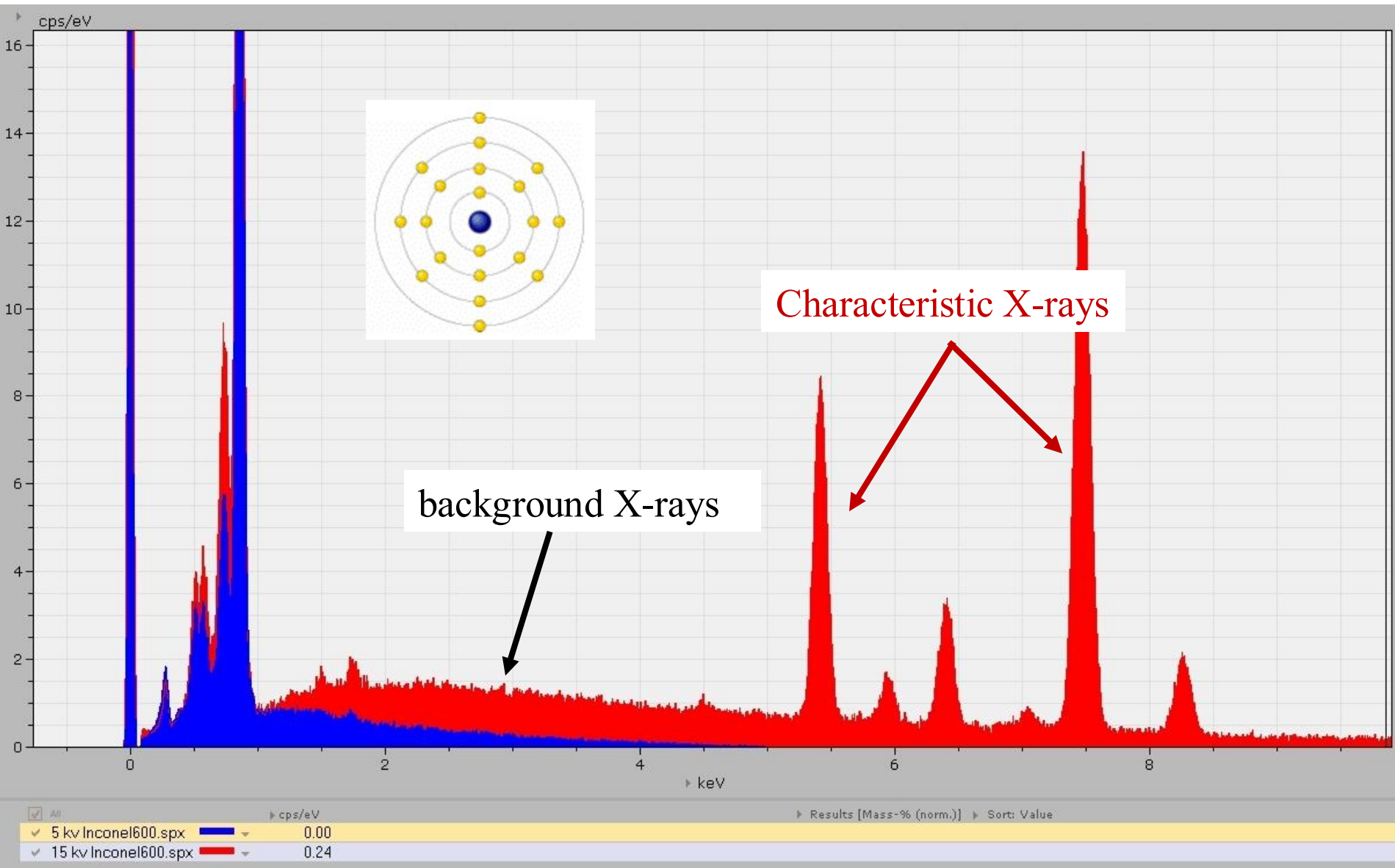
Four blue arrows point to the variables in the equation with their respective units:

- atomic weight [g/mol] points to A
- energy of beam [keV] points to E_o
- atomic number points to Z
- density g/cm³ points to ρ

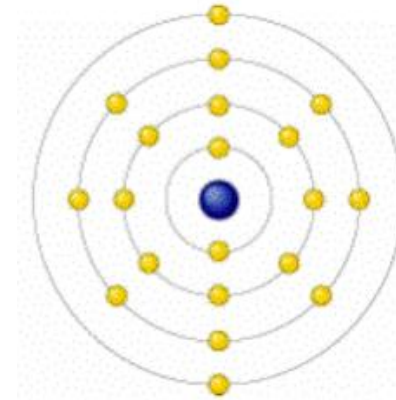
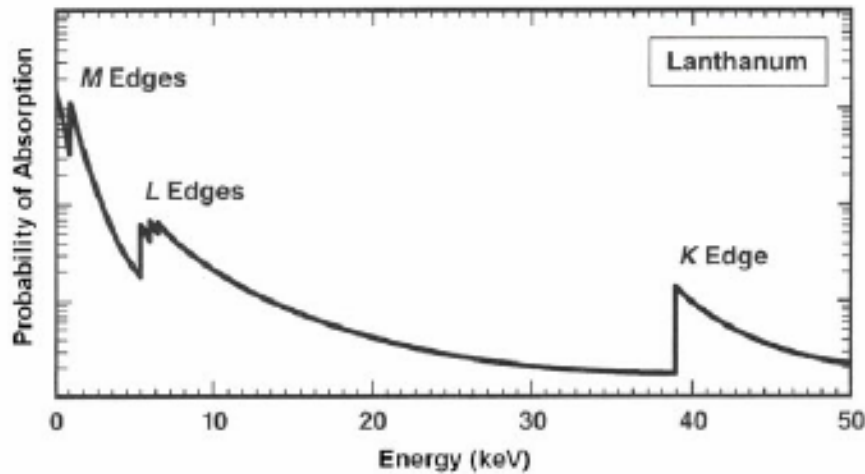
On the right, a schematic diagram shows a horizontal line separating the "vacuum" region above from the "sample" region below. A teardrop-shaped dashed line within the sample represents the "electron interaction volume". A vertical double-headed arrow to the right of this volume is labeled R_e , indicating the depth of the interaction.

- a) To detect Pt in the sample, R_e must be greater than the energy of x-rays emitted by Pt atoms.
- b) The quantities R_e and E_o are not relevant to x-ray analysis.
- c) To detect all elements that are present in the sample, the energy E_o must be greater than the maximum energy of characteristic x-rays emitted by Si, O, Pt and Fe atoms.
- d) To detect Pt in the sample E must be smaller the energy of x-rays emitted by Pt atoms.

EDX of steel: 5keV (Blue) & 15 keV (red)



X-ray absorption: Sample test question



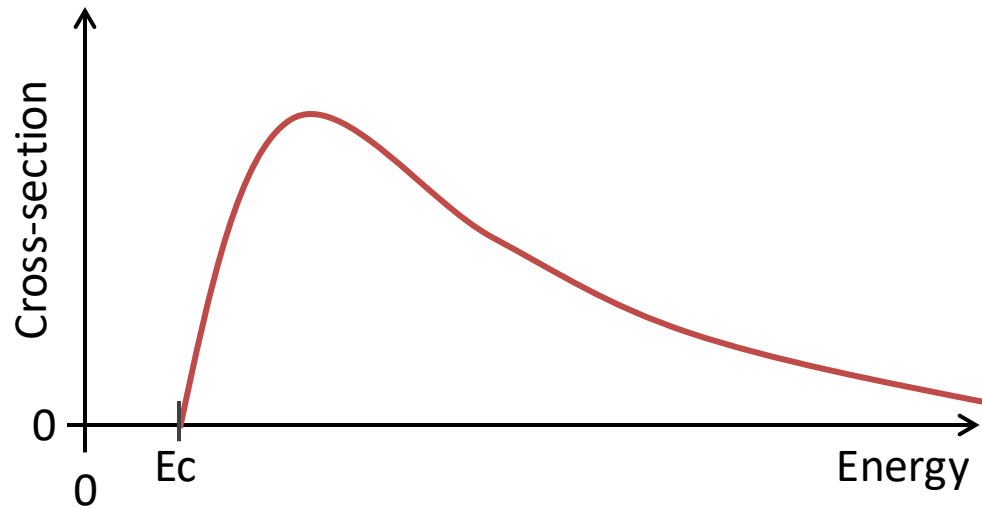
This x-ray absorption spectrum shows a number of so-called x-ray “absorption edges” for the element lanthanum. The edges are labelled K, L and M.

The K absorption edge is at energy E_a and the K x-ray has energy E_x .

Which statement is true?

- a) E_a is equal to the energy of Auger electrons emitted from lanthanum atoms.
- b) E_a is equal to E_x .
- c) E_a greater than E_x .
- d) E_a is smaller than E_x .

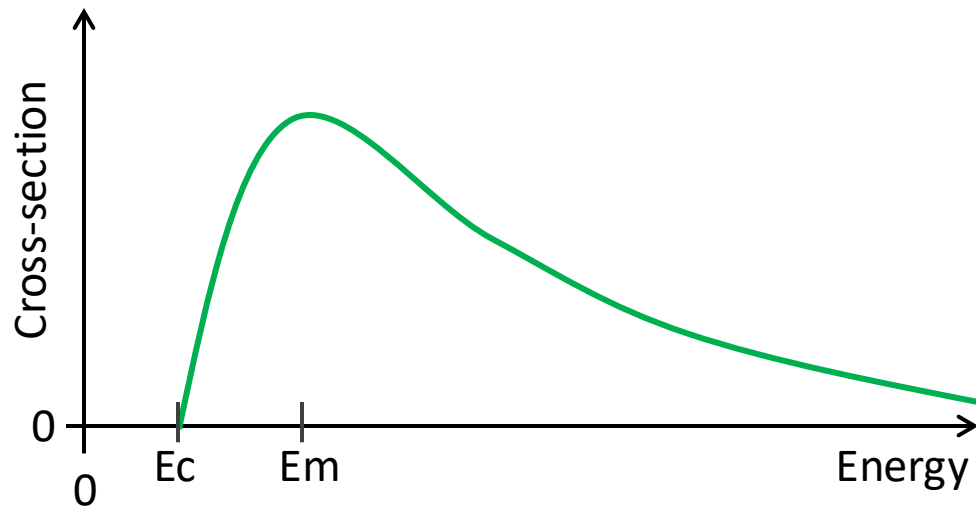
Cross-sections



The graph shows a cross-section that pertains to x-ray analysis in an electron microscope. A cross-section has units of area and it represents a probability, expressed here as a function of energy. Which statement is true?

- a) A cross-section can not equal to zero below the energy E_c . This is physically impossible and hence this plot must be incorrect.
- b) This is a cross-section for an electron relaxation transition that gives rise to x-ray emission.
- c) This is a cross-section for ionisation of an atom by an electron.
- d) The quantity on the horizontal axis is the energy of the emitted x-ray.

Scattering cross-section



The graph shows a cross-section plotted as a function of electron energy. The general shape of this curve is typical of some electron collision processes (i.e., as the energy increases from zero, the cross section is zero up to E_c , then increases up to a maximum at E_m and then decreases at energies greater than E_m). What is the quantity E_c ?

- a) Binding energy of a particle involved in the collision.
- b) Kinetic energy of a particle involved in the collision.
- c) E_c is not a real quantity. This plot doesn't make sense because a cross-section must be a non-zero value at all energies greater than zero.

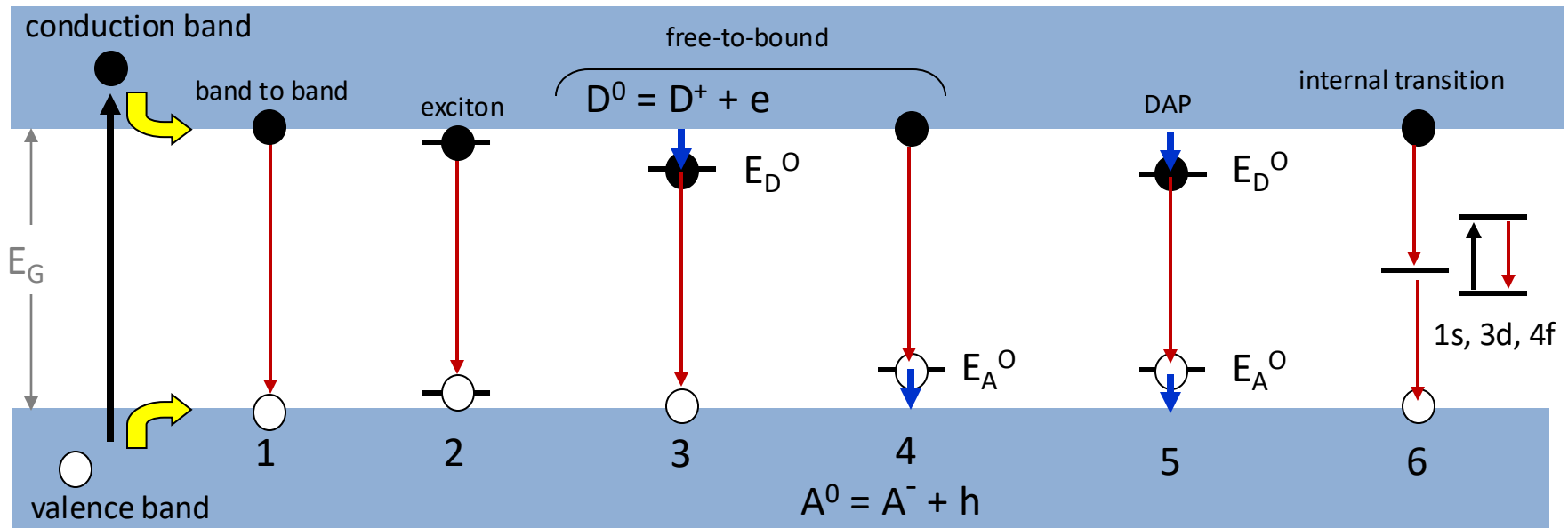
Ion-solid interactions

Consider a piece of single crystal silicon (Si) irradiated by a focused argon (Ar^+) ion beam. The ions generate crystal damage (i.e., defects) in the Si. The defect generation rate can be expressed as a “density per unit time”, denoted by the symbol D , in units of [defects per nm^3 per second].

Which of the following statements is true?

- a) Increasing the ion beam energy (and keeping all else fixed) will reduce the ion penetration range in the Si sample.
- b) Changing the ion beam species from Ar^+ to Xe^+ (and keeping all else fixed) will reduce D because the penetration range of Xe^+ is lower than that of Ar^+ .
- c) Changing the ion beam species from Ar^+ to He^+ (and keeping all else fixed) will reduce D because the penetration range of He^+ is lower than that of Ar^+ .
- d) Changing the ion beam species from Ar^+ to Xe^+ (and keeping all else fixed) will increase D because Xe^+ ions are heavier than Ar^+ ions.

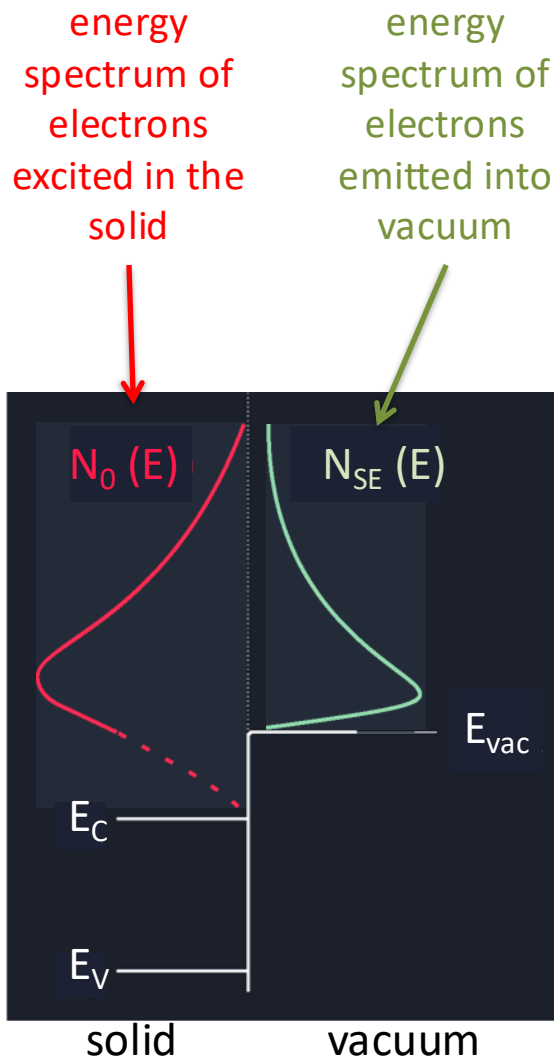
Cathodoluminescence



Consider this electron energy diagram showing various electron transitions in a sample measured using cathodoluminescence (CL) spectroscopy. Which statement is true?

- a) The black arrows indicate electron excitation pathways in a metal.
- b) The red arrows indicate electron excitation pathways in a metal.
- c) The red arrows indicate electron excitation pathways in a semiconductor.
- d) The red arrows indicate electron excitation pathways in a dielectric.
- e) The black arrows indicate photon absorption pathways in a dielectric.
- f) The black arrows indicate photon absorption pathways in a metal.

Sample class test question

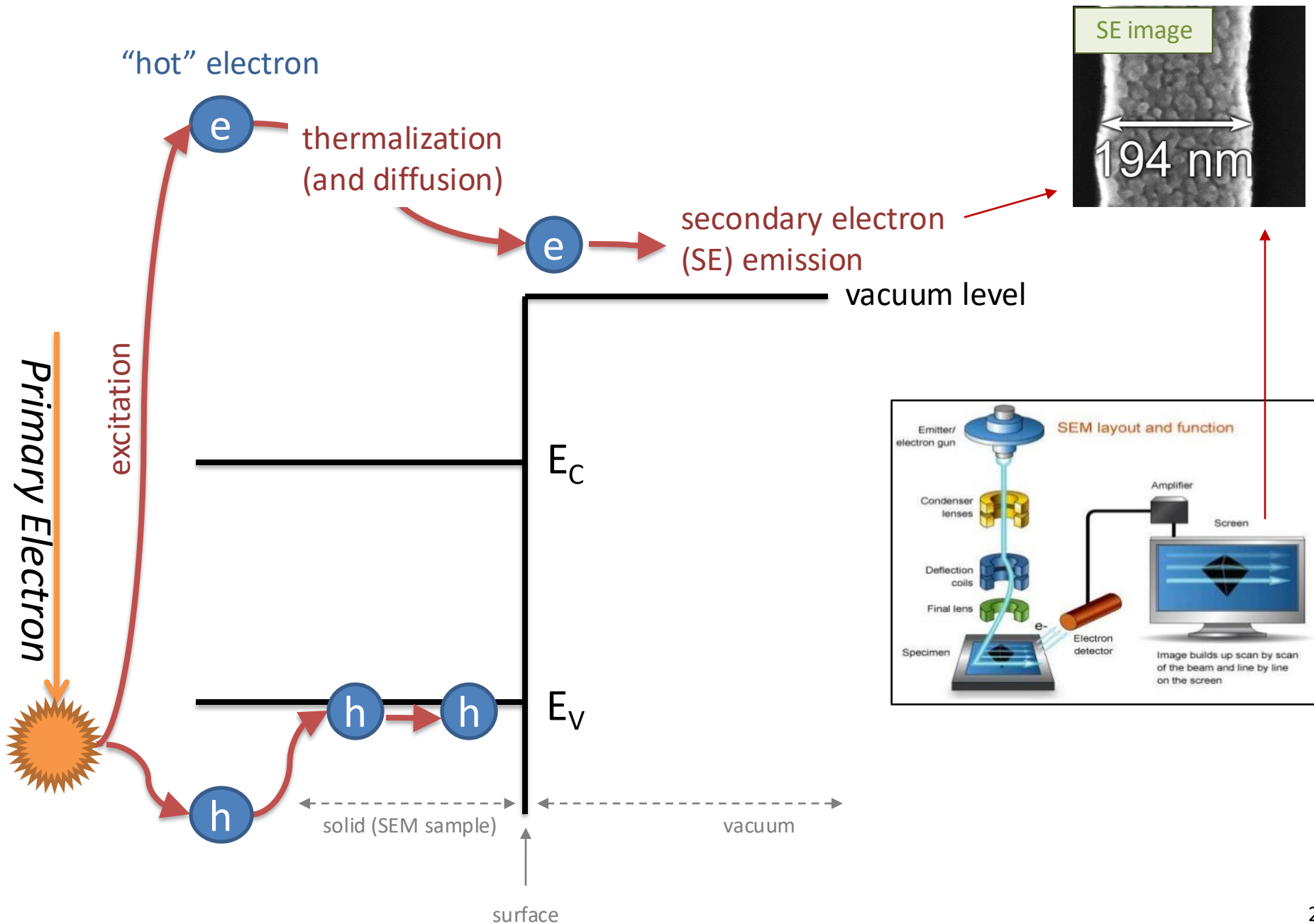


Consider this electron energy diagram of the sample-vacuum interface. Superimposed on the diagram are plots of the energy spectra of secondary electrons excited in the sample by an electron beam, and secondary electrons emitted from the sample and used to form an image.

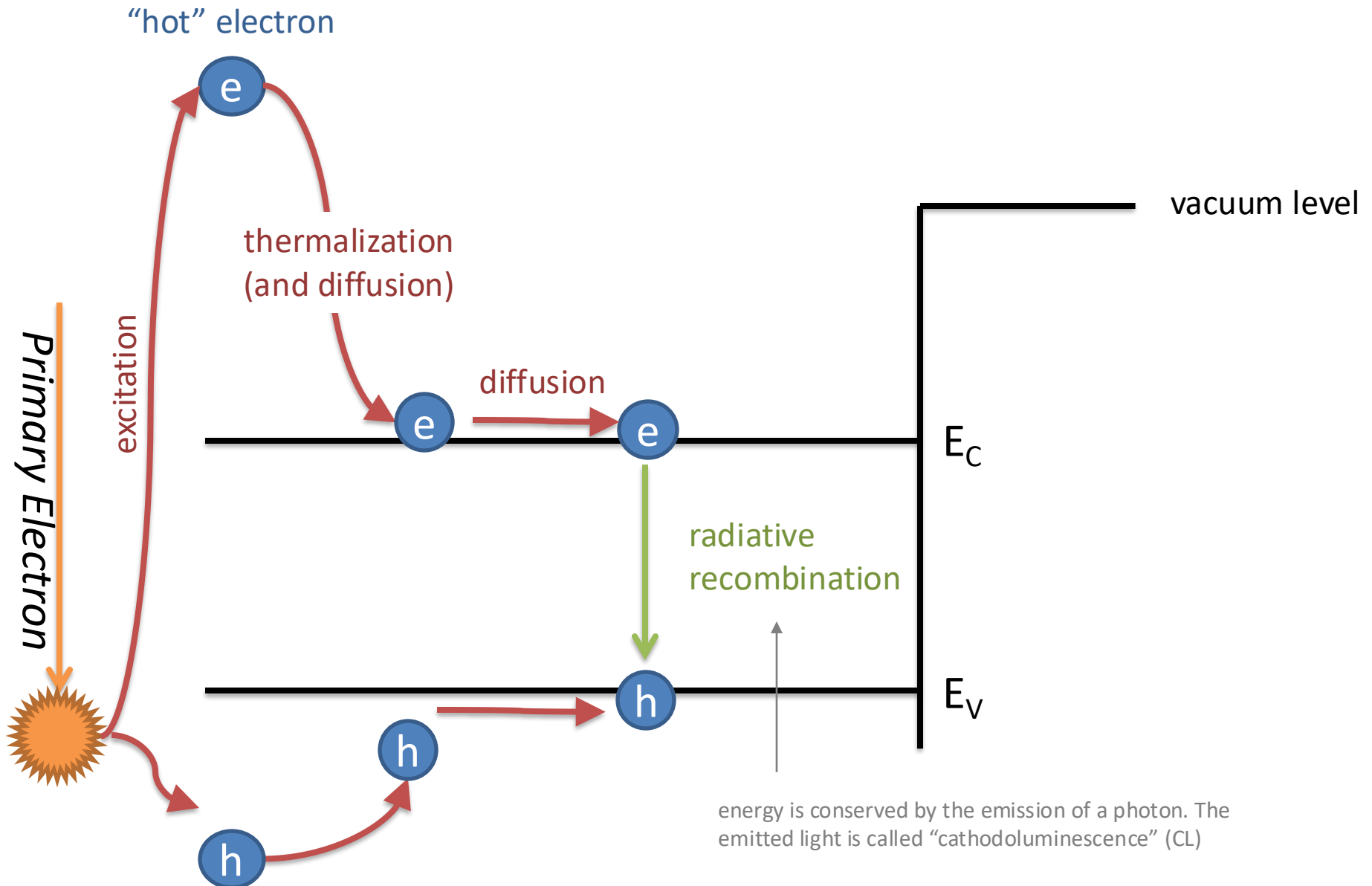
Why are the two energy spectra not identical?

- a) Because emitted secondary electrons are accelerated towards a detector.
- b) Because of electron-hole pair recombination in the sample.
- c) Because of electron diffusion in the sample.
- d) Because the electron image is out of focus.
- e) Because the electron beam energy is smaller than the bandgap energy.

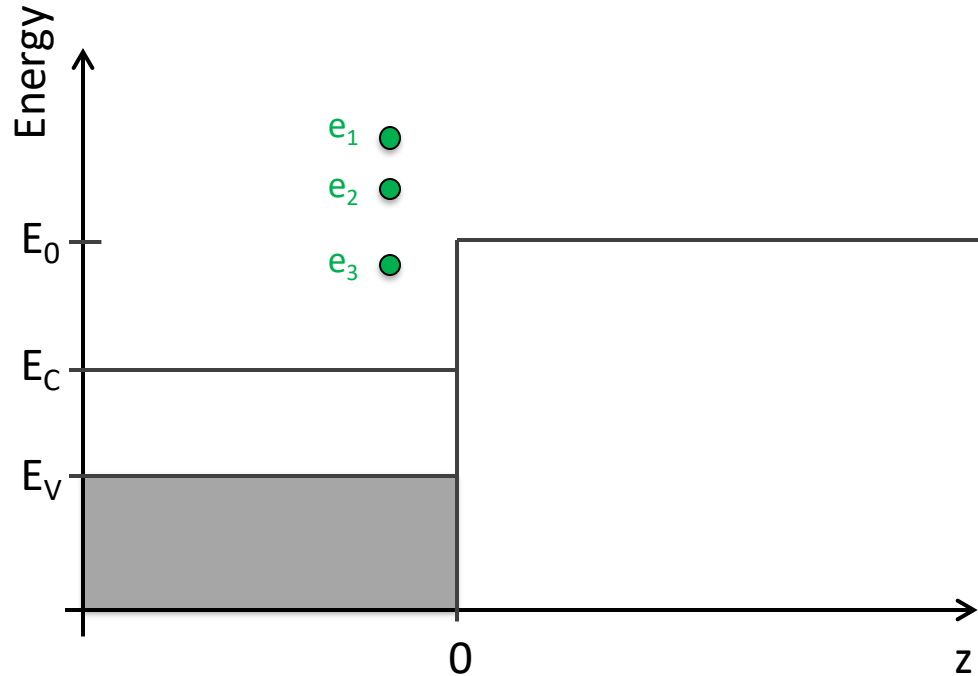
SEM: Electron excitation & emission



CL generation



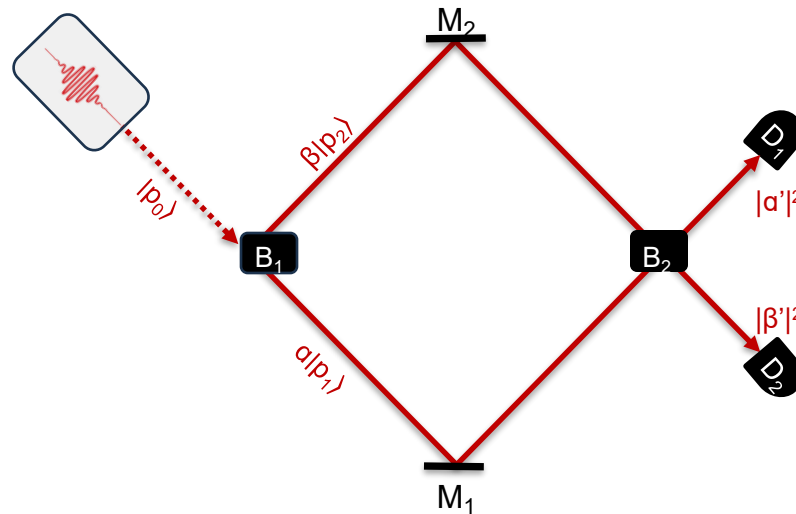
Secondary electron emission



Consider this electron energy diagram of a sample imaged using a scanning electron microscope. The circles labelled e_1 , e_2 and e_3 represent electrons excited in the sample by the electron beam. Your lecturer makes the claim that “electron e_1 is guaranteed to be emitted from the sample with a probability of 100%”. This statement is TRUE or FALSE because:

- a) FALSE because the electron energy is too low.
- b) FALSE because the electron energy is too high.
- c) TRUE because the electron energy is greater than the vacuum level.
- d) TRUE because the electron energy is greater than the work function of the sample.
- e) TRUE because energy must be conserved when the electron crosses the solid-vacuum interface.
- f) TRUE because because momentum must be conserved when the electron crosses the solid-vacuum interface.
- g) FALSE because because momentum must be conserved when the electron crosses the solid-vacuum interface.

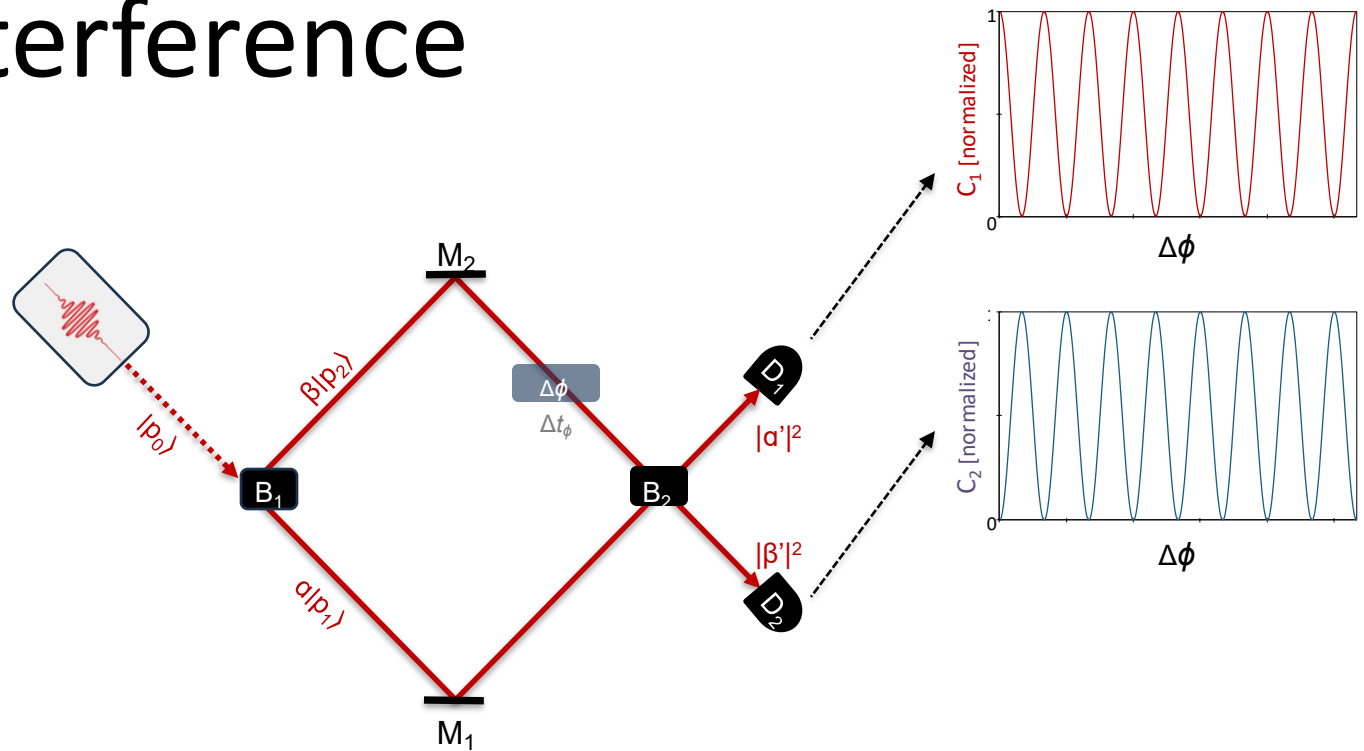
Photon interference



Consider this single photon interference setup. M, B and D represent mirrors, 50%-50% beamsplitters and detectors. p_1 and p_2 represent 2 possible photon paths in the setup. Which statement is true?

- a) Each photon injected into the setup splits into 2 photons at B_1 . The 2 photons subsequently recombine and form a single photon at B_2 .
- b) Each photon injected into the setup has a 50% chance of travelling to detector D_1 along path 1 and 50% chance of travelling to detector D_2 along path 2.
- c) The wave function of each photon injected into the setup splits into 2 components at B_1 , which explore paths 1 and 2 in parallel in a superposition state which collapses into a definite state at B_2 .
- d) The wave function of each photon injected into the setup splits into 2 components at B_1 , which explore paths 1 and 2 in parallel in a superposition state which collapses into a definite state at one of the detectors.

Photon interference



Consider this single photon interference setup. M, B and D represent mirrors, 50%-50% beamsplitters and detectors. C_1 and C_2 are photon detection rates at detectors 1 and 2, measured versus the phase shift $\Delta\phi$. Which statement is true?

- C_1 and C_2 are interference patterns generated because each photon injected into the setup self-interferes at D_1 and D_2 .
- C_1 and C_2 are interference patterns generated because each photon injected into the setup self-interferes at B_1 .
- C_1 and C_2 are interference patterns generated because each photon injected into the setup self-interferes at B_2 .
- C_1 and C_2 are interference patterns generated because consecutive photons injected into the setup interfere at B_1 .
- C_1 and C_2 are interference patterns generated because consecutive photons injected into the setup interfere at B_2 .